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Method and arrangement for processing a signal

Abstract

A wireless communication unit to facilitate determining a plurality of communication link cost metrics at a receiving wireless communication unit using a multiple modulation and coding scheme, MCS, in a wireless mesh network, is described. A transmitter transmits sounding transmissions to at least two neighboring wireless communication units. A processor generates a sounding burst comprising: a plurality of multicast sounding transmissions that employ a different MCS; and a plurality of parameters contained at an application layer protocol level. The plurality of parameters comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a particular table within a set of tables which is then used to determine a transmission bit rate, B, at the receiving unit for each multicast sounding transmission within the sounding burst.

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Chorobczek, J. et al., 'I Hear You' (IHU) message (see babel RFC6126/RFC 8966), <https://www.rfc-editor.org/rfc/rfc8966.html>, Jan. 2021 (45 Pages). cited by applicant

Draves, Richard et al., 'Routing in Multi-Radio, Multi-Hop Wireless Mesh Networks', MobiCom'04, Sep. 26-Oct. 1. 2004, Philadelphia, Pennsylvania, USA, <https://www.cs.montana.edu/courses/spring2009/541/Ref/Draves-MobiCom04.pdf> (15 pages). cited by applicant

Bicket, John et al., 'Architecture and Evaluation of an Unplanned 802.11b Mesh Network', MobiCom'05, Aug. 28-Sep. 2, 2005, Cologne, Germany, <https://pdos.csail.mit.edu/papers/roofnet/mobicom05/roofnet-mobicom05.pdf> (12 pages). cited by applicant

The new sub Type Length Value, TLV, encoding sounding transmission, Wikipedia the page was last edited on Feb. 4, 2023, [https://en.wikipedia.org/wiki/Type%E2%80%93length%E2%80%93value#:~:text=Within%20communication%20protocols%2C%20TLV%20\(type,and%20finally%20the%20value%20itself](https://en.wikipedia.org/wiki/Type%E2%80%93length%E2%80%93value#:~:text=Within%20communication%20protocols%2C%20TLV%20(type,and%20finally%20the%20value%20itself) (3 pages). cited by applicant

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Background/Summary

RELATED APPLICATION (1) This application claims the benefit of U.S. Provisional application No. 63/282,431, filed Nov. 23, 2022, which is incorporated by reference in its entirety herein.

TECHNICAL FIELD

(1) The field of this invention relates generally to wireless communication technologies that employ a multi-modulation and coding scheme (MCS). In particular, the field of this invention relates to wireless mesh 802.11 WiFi™ networks and specifically a form of communication link estimate transmission time (ETT) cost metric that is often used in routing protocols used in MCS systems such as 802.11 WiFi™ networks.

BACKGROUND

(2) A recent development in fourth generation (4G) and fifth generation (5G) wireless communications is the use of wireless mesh networks. Both of these technologies are compliant with third generation partnership project (3GPP™) standards.

this example, there are three possible links: A direct link **115** between node **5 150** and node **1 110**. This single link has a loss rate of 0.8; A two-hop link **125** using node **2 120**. Each link in this case has a loss rate of 0.2; A three-hop link **135** using node **3 130** and node **4 140**. Each link in this case has a loss rate of 0.

(14) A simple 'hop count' cost metric would choose the direct path, i.e., link **1 115** above. As determined in equation [1] above, it would take an expected five transmissions to reach node **1 110**. However, if the ETX metric was employed, then the cost metric calculations for each possible link are: case (i) $ETX=5$, case (ii) $ETX=1.25+1.25=2.5$, case (iii) $ETX=1+1+1=3$

(15) Thus, using the ETX metric the 2-hop path **125** via node **2 120** is selected requiring on average 2.5 transmissions to reach node **1 110**.

(16) Many wireless technologies (notably including 802.11 WiFi) employ a rate adaptation mechanism, by employing multiple modulation and coding schemes (MCS). This feature results in a problem associated with the ETX metric. As discussed above the link loss rate is effectively measured for the lowest rate within the set of allowed modulation and coding schemes. This will result in inefficient routes being selected as the routing metric will recommend a route based on the assumption of lowest MCS when other routes using higher rate MCS could be employed. Thus, for example using the ETX metric, routes selected would tend to employ a small number of long hops, which could only support a low MCS, as opposed to another possible route involving a larger number of shorter hops that employs a high MCS. So, for example ETX might pick a single hop that could support only 1 Mbps, but another route employing two hops at 10 Mbps may also exist and could provide better overall performance.

(17) These problems, to some degree, were addressed in the development of the Estimate Transmission Time (ETT) cost metric, which is described in the document 'Routing in Multi-Radio, Multi-Hop Wireless Mesh Networks', authored by: Richard Draves, Jitendra Padhye, Brian Zill (<https://www.cs.montana.edu/courses/spring2009/541/Ref/Draves-MobiCom04.pdf>).

(18) This ETT cost metric takes into account both the number of re-transmissions and the transmission bandwidth or bit rate (where the reference uses the term 'bandwidth B') in the communication link, using both of these two parameters to define how long it will take to transmit a packet. This metric is defined as [3]:

$$(19) \text{ cost_ETT} = \frac{1}{1-p} * \frac{S}{B} = \text{cost_ETX} * \frac{S}{B} \quad [3]$$

where S is the size of the packet that is to be sent and B is the available bandwidth or bit rate on the link.

(20) Using this ETT cost

required for all mesh nodes to have knowledge (at a higher protocol layer) of the MCSs that are to be used in the sounding process.

(36) As mentioned above a number of versions of the 802.11 standard are now typically implemented, e.g., 802.11n, 802.11ac and 802.11ax. Table 1 illustrates a standard MCS rate table for 802.11n. Table 2 illustrates a standard MCS rate table for 802.11ac. Table 3 illustrates a standard MCS rate table for 802.11ax. The inventors have recognized and appreciated that a mechanism is required by which all these versions can be supported easily when determining the ETT parameter.

(37) Thus, there exists a need to provide an improved routing selection mechanism, particularly for wireless mesh networks that are compliant with 802.11 WiFi networks.

SUMMARY OF THE INVENTION

(38) In a first aspect of the invention, a wireless communication unit for determining a plurality of link costs for communications to a receiving wireless communication unit using a multiple modulation and coding scheme, MCS in a wireless mesh network, is described. A transmitter transmits packets to at least two neighboring wireless communication units of a plurality of neighboring wireless communications units. A processor generates a sounding burst comprising: a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS to be transmitted to the plurality of neighboring wireless communication units; and a plurality of parameters contained at an application layer protocol level. The plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of packets in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst.

(39) In this manner, embodiments of the invention may allow a use of the ETT metric within a population of mesh nodes that employs a diversity of modern versions of 802.11 e.g., 802.11n and 802.11ac. These have many different MCS and many different implementation options affecting the MCS that they can support. This approach is in contrast to known techniques whereby there is no current mechanism (or possibility) for the mesh nodes to obtain a full understanding of the MCSs that are being used in the sounding bursts, and therefore makes it impossible to employ the ETT metric in a routing algorithm. In this manner, examples described herein further allow ETT metrics to be determined when multiple 802.11 versions are used (together with multiple options within the versions). Furthermore, examples described herein provide an efficient means for sounding histories over multiple MCS to be maintained. Additionally, examples described herein may remove a need to know a sounding transmission packet size when determining ETT metrics.

(40) In an optional example, a mechanism is provided that ties together whether certain sounding transmissions have been received at which MCS, which may be performed at the application layer. In this manner, having this information directly available to the routing protocol via the application layer, simplifies functionality considerably and enables a direct link between a higher layer packet and what MCS was used at the physical layer. Ensuring that parameters needed to correctly measure link cost, i.e., the parameters contained in the sounding burst as described herein, are available at the same protocol layer as the routing protocol ensures that all functionality is contained in one protocol layer and ensures that cross layer functionality is not needed. Note that in many implementations programming access to physical layer is not possible and even in cases where it is tying together MCS used for transmissions along with appropriate acknowledge (ACK)/no acknowledge (NACK) reports is extremely challenging.

burst for each of the received sounding transmissions received via the plurality of communication links.

(47) In an optional example, the processor may be configured to calculate an estimate transmission time (ETT) cost metric for the respective received sounding burst for each of the received sounding transmissions received via the plurality of communication links using a relative value between a peak bit rate and an individual bit rate contained in a MCS rate table.

(48) In an optional example, the processor may be further configured to use a relative MCS rate table index contained in the sounding burst together with predefined rate table information to obtain a MCS rate cost metric for all MCS used in sounding burst. In an optional example, the processor may be further configured to derive a delivery rate from the parameters contained in the sounding transmission and use both the derived delivery rate and a MCS rate to derive a plurality of ETT cost metric values for each MCS used in the sounding burst.

(49) In an optional example, the processor may be further configured to construct a sounding history of transmissions from the transmitting wireless communication unit received via the plurality of communication links using parameters in sounding burst. In an optional example, the processor may be further configured to derive a packet delivery rate for each MCS for each of the received sounding transmissions received via the plurality of communication links. In an optional example, the processor is coupled to a transmitter in the receiving wireless communication unit and the calculated cost metric is transmitted to at least one neighboring wireless communication unit for use by the neighboring wireless communication unit with a conventional routing algorithm to determine a performance for each of a plurality of routes through the wireless mesh network. In an optional example, the wireless mesh network may comply with at least one of: a WiFi™ 802.11n, a WiFi™ 802.11ac, a WiFi™ 802.11ax standard.

(50) In a fourth aspect of the invention, a method for determining a plurality of link costs for communications in a wireless mesh network received at a receiving wireless communication unit configured to use a multiple modulation and coding scheme, MCS, is described. The method comprising: receiving sounding transmissions from at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between a transmitting wireless communication unit and the receiving wireless communication unit; and receiving and processing a sounding burst comprising a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS from respective neighboring wireless communication units of the plurality of neighboring wireless communication units; and each multicast sounding transmission comprises a plurality of parameters contained at an application layer protocol level. The plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst. The method further comprises calculating a cost metric for the respective received sounding burst for each of the received sounding transmissions received via the plurality of communication links.

(51) In a fifth aspect of the invention, a communication system comprises a wireless communication unit according to the first aspect and a receiving wireless communication unit according to the third aspect.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further details, aspects and embodiments of the invention will be described, by way of example only, with reference to the drawings. In the drawings, like reference numbers are used to identify like or functionally similar elements. Elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale.
- (2) FIG. 1 illustrates an overview of a known wireless communication system that uses ETX measurements and a loss rate for different possible MCS to provide a benefit over a simple hop count metric in wireless mesh networks.
- (3) FIG. 2 illustrates a known basic sounding functionality associated with an ETX metric determination.
- (4) FIG. 3 illustrates a known sounding mechanism for multi rate sounding in 802.11b that sends a sequence of beacon transmissions at multiple different bit rates.
- (5) FIG. 4 illustrates an overview of a wireless communication system configured to provide an improved routing

(22) Skilled artisans will appreciate that elements in the figures are illustrated with simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions and/or relative positioning of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of various embodiments of the present invention. Also, common but well-understood elements that are useful or necessary in a commercially feasible embodiment are often not depicted in order to facilitate a less obstructed view of these various embodiments of the present invention. It will further be appreciated that certain actions and/or steps may be described or depicted in a particular order of occurrence while those skilled in the art will understand that such specificity with respect to sequence is not actually required. It will also be understood that the terms and expressions used herein have the ordinary technical meaning as is accorded to such terms and expressions by persons skilled in the technical field as set forth above except where different specific meanings have otherwise been set forth herein.

DETAILED DESCRIPTION

(23) Examples of the invention propose an improved routing selection mechanism, particularly for wireless mesh networks that are compliant with 802.11 WiFi networks. Examples of the invention describe a wireless communication unit for determining a plurality of link costs for communications to a receiving wireless communication unit using a multiple modulation and coding scheme, MCS in a wireless mesh network. A transmitter transmits sounding transmissions to at least two neighboring wireless communication units of a plurality of neighboring wireless communications units. A processor generates a sounding burst comprising: a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS to be transmitted to the plurality of neighboring wireless communication units; and a plurality of parameters contained at an application layer protocol level. The plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst. In this manner, the proposed mechanism is able to ensure that all mesh nodes have knowledge (at a higher protocol layer, such as the application layer) of the MCSs that are to be used in the sounding process.

(24) In the context of the description, the term ‘wireless communication unit’ encompasses any wireless device that is able to be configured to support communications with one or more further wireless communication units in a wireless mesh network.

(25) Although examples of the invention have been described with reference to 802.11 physical layer standards, it is envisaged that the examples herein described may be employed in any wireless communication system supports multi-MCS operation and can provide a means to support multicast functionality at all available MCS, which would result in a MCS rate table index pointing to a technology different from 802.11.

(26) Referring now to FIG. 4, a wireless communication system 400 is shown in outline, in accordance with one example embodiment of the invention. In this example embodiment, the wireless communication system 400 is compliant with, and contains network elements capable of operating over, a universal mobile telecommunication system (UMTS™) air-interface. In particular, the embodiment relates to a system's architecture for an Evolved-UMTS Terrestrial Radio Access Network (E-UTRAN) wireless communication system, which is currently under discussion in the third Generation Partnership Project (3GPP™) specification for long term evolution (LTE), based around OFDMA (Orthogonal Frequency Division Multiple Access) in the down

(35) As illustrated, the EPS **404** is operably connected to two eNodeBs **410**, with their respective coverage zones or cells **490** and a plurality of wireless mesh nodes **425** receiving transmissions from the EPS **404** via the eNodeBs **410**. In accordance with example embodiments of the present invention, at least one eNodeB **410** and at least one wireless mesh node **425** (amongst other elements) have been adapted to support the concepts hereinafter described.

(36) The main component of the RAN is an eNodeB (an evolved NodeB) **410**, which performs many standard base station functions and is connected to the EPS **404** via an S1 interface and to the wireless mesh nodes **425** via a Uu interface. A wireless communication system will typically have a large number of such infrastructure elements where, for clarity purposes, only a limited number are shown in FIG. 4. The eNodeBs **410** control and manage the radio resource related functions for a plurality of wireless communication units/terminals (such as wireless mesh node **425**). Each of the wireless mesh nodes **425** comprises at least a transceiver unit **427** operably coupled to signal processing logic (with one wireless mesh node illustrated in such detail for clarity purposes only). An example wireless mesh node **425** is illustrated in greater detail in FIG. 5. The system comprises many other wireless mesh nodes **425** and eNodeBs **410**, which for clarity purposes are not shown.

(37) In examples of the invention, a number of wireless mesh nodes **425** have been adapted to include additional information within each transmitted WiFi™ message **470**, **471**, **472** of a sounding burst in order to provide the receiving wireless communication unit (e.g., another wireless mesh node) the ability to easily determine whether (or not) an individual sounding transmission within the sounding burst has been received. Furthermore, the additional information within each transmitted WiFi™ message **470**, **471**, **472** of a sounding burst has been adapted to inform the receiving wireless communication unit (e.g., another wireless mesh node) with the ability to determine what version of 802.11 is being used by the transmitter (e.g., 802.11n, 802.11ac or 802.11ax). Furthermore, the additional information within each transmitted WiFi™ message **470**, **471**, **472** of a sounding burst has been adapted to inform the receiving wireless communication unit (e.g., another wireless mesh node) with the ability to determine the exact MCS at which each of the sounding transmissions in the sounding burst is using. Furthermore, the additional information within each transmitted WiFi™ message **470**, **471**, **472** of a sounding burst has been adapted to inform the receiving wireless communication unit (e.g., another wireless mesh node) with the ability to simplify the task of finding the appropriate rate value to use in the MCS tables.

(38) In this manner, the receiving wireless communication unit is not dependent on 802.11 options, such as guard interval and the transmission bit rate, B, as shown in Tables 1 to 3, which show the case for 802.11n, 802.11ac and 802.11ax. Tables 1 to 3 contain all possible MCS, together with the associated bit rate, notably containing all the possible spatial streams, up to '4' in the case of 802.11n and up to '8' in the case of 802.11ac. However, typical 802.11 implementations have fewer than this, typically only '2'. In contrast, as illustrated in Table 4 an improved updated MCS rate table for 802.11n with relative ETT cost metric values can be generated and distributed. Similarly, as illustrated in Table 5

(46) The cost metric calculator **544** takes the costs reported from the beacon RX circuit **534** (representing an RX cost) and the “I hear you” (IHU) processing circuit **532** (representing a TX cost) and processes this to form a single ETT cost metric for each of the neighbors of the current mesh node. The ETT cost metric is passed to the IP tables/routing circuit **540**.

(47) The IP tables/routing circuit **540**, using a conventional routing protocol such as RIP, OSPF or babel, employs the ETT cost metric to find the best route through the mesh. This information is configured with IP route tables, which is passed on the IP stack circuit **538**. The IP stack circuit **538** is a conventional IP stack, which is used to send data over the mesh routing is controlled by the IP tables/routing circuit **540**.

(48) In order to achieve these benefits, a processor in the transmitting communication unit includes the following parameters in each sounding transmission within the sounding burst by the respective transmitting wireless communication unit, e.g., wireless mesh node **425**. First, an intra-burst sequence number is added (in addition to the conventional sequence number). In examples of the invention, the respective transmitting wireless communication unit increments the intra-burst sequence number for each transmission within the sounding burst.

(49) Secondly, information on a number of sounding transmissions used in the sounding burst is added to the sounding transmission. In some examples, a further parameter of an indication of how many individual sounding transmissions are to be used in the sounding burst may be included. In some examples, this is contained in each individual sounding transmission in the sounding burst and is the same value within each of these sounding transmissions. In some examples, an enhancement to this functionality may be implemented whereby a non-consecutive MCS is deployed in the burst, as later described.

(50) Thirdly, an additional parameter of a MCS rate table index may be included in the sounding transmission within the sounding burst by the respective transmitting wireless communication unit, e.g., wireless mesh node **425**. In some examples, this parameter may be a pointer to a previously defined table which is a simplified version of the 802.11 rate tables (see tables 1 to 3 for the original tables). Note that typically there is a separate table for each version of 802.11 (i.e., one for 802.11n, another for 802.11ac, etc.). In this manner, the node/receiving wireless communication unit can then use a simple lookup with the appropriate table to help calculate the ETT metric

(51) The signal processor **428** in the transmit chain may be implemented as distinct from the signal processor in the receive chain. Alternatively, a single processor may be used to implement a processing of both transmit and receive signals, as shown in FIG. 5. Clearly, the various components within the wireless communication unit **425** can be realized in discrete or integrated component form, with an ultimate structure therefore being an application-specific or design selection.

(52) Examples of the generated sounding burst comprising a plurality of sounding multicast transmissions to be transmitted to any receiving (neighboring) wireless communication unit, wherein the sounding burst comprises a plurality of additional parameters, is shown in FIGS. **6**, **7** and **8**. FIG. **6** illustrates a generated sounding burst comprising a plurality of sounding transmissions to be transmitted to at least one neighboring wireless communication unit, where the generated sounding burst is configured for, say, a scenario employing an 802.11n (with 256 QAM support at 5 GHz) single spatial stream **600** and comprises a plurality of additional parameters defined in the higher layer message, adapted in accordance with some

set of possible cases) and inside the 802.11n version there are a large number of options which affect the bit rate, hence the large size of the tables. If the ETT parameter was calculated directly as described in equation 2 then it is necessary to know all of these parameters plus the size of the sounding transmission. This is clearly a potentially complex calculation particularly informing neighbor nodes about sounding transmission size as this may change over the course of the connection.

(64) However, the key issue regarding cost calculation is the relative value both within a single link between different MCS and between different links across the network as was shown in the simple examples **100**, **200** shown in FIGS. **1** and **2**. This leads to the following enhancements:

(65) Rather than use absolute values for the MCS bit rate determination, a relative number can be obtained based on the peak bit rate obtainable, for example the processor calculates a cost metric, such as an estimate transmission time (ETT) cost metric, for a respective received sounding burst for each of the received sounding transmissions received via the plurality of communication links using a relative value between a peak bit rate and an individual bit rate contained in a MCS rate table.

(66) It is not necessary to use an absolute value for the packet size as this will be the same for all links investigated. It will almost always be 1500 bytes for the typical case of sending TCP data to a neighboring node (maximum packet size) and because this is unchanging it can be removed from the calculation.

(67) Looking at the relative bit rate issue in more detail and taking the example of 802.11n with 20 MHz bandwidth and 0.8 mS guard interval (GI), the lowest bit rate, associated with MCS1/single spatial stream, is 6.5 Mbps and the highest bit rate, associated with MCS31/four spatial streams, is 260 Mbps (see table 1). The ratio of these is thus:

260 Mbps/6.5 Mbps=40

(68) Thus, the relative cost of MCS1 over MCS31 is 40 and it would be possible to assign a cost metric of 1 to MCS31 and 40 to MCS1. This can be repeated for all possible MCS in 802.11n (20 MHz bandwidth/0.8 mS) such that for example MCS2 would be assigned an 'MCS rate cost' of 260/13=20 etc.

(69) Additionally, it is notable that for all the other possible combinations of bandwidth and guard interval (i.e., 20 MHz BW/0.4 mS GI, 40 MHz BW/0.8 mS GI and 40 MHz BW/0.4 mS) the same MCS rate costs apply. For example, for the case of 40 MHz BW/0.4 mS GI MCS1 is associated with a rate of 15 Mbps and MCS31 is associated with a rate of 600 Mbps thus the ratio between these is again 40

(70) Thus, it is observed that for the case of 802.11n table 1 can be simplified to a simple lookup of MCS to MCS rate cost. This simplified table is shown in table 4. Similarly, the MCS rate cost tables for 802.11ac and 802.11ax are shown in tables 5 and 6 respectively.

transmissions are received over the window period therefore $PDR.sub.MCS4=5/6=0.8333$. Whereas, for MCS5: 3 out of 6 sounding transmissions are received over the window period therefore $PDR.sub.MCS5=3/6=0.5$. Whereas for MCS6 and MCS7 no sounding transmissions were received over the window period therefore $PDR.sub.MCS6=0$ and $PDR.sub.MCS7=0$.

(89) $MCS_rate_cost.sub.MCS$

(90) Here, $MCS_rate_cost.sub.MCS_at$ **930** may be obtained via simple lookup. The MCS rate table index in this example is '0'. Table 4 illustrates the corresponding table for this index value. Reading from this table, for the corresponding MCS in this example (MCS 0, 1, 2, 3, 4, 5, 6, 7), results in the following MCS_rate_cost :

(91) $MCS_rate_cost.sub.MCS0=40$, $MCS_rate_cost.sub.MCS1=20$, $MCS_rate_cost.sub.MCS2=13.333$, $MCS_rate_cost.sub.MCS3=10$, $MCS_rate_cost.sub.MCS4=6.6667$, $MCS_rate_cost.sub.MCS5=5$, $MCS_rate_cost.sub.MCS6=4.444$, $MCS_rate_cost.sub.MCS7=4$.

(92) $Cost_ETT.sub.MCS$

(93) Using equation 4 results in the following values for $Cost_ETT.sub.MCS$ at **940**:

$Cost_ETT.sub.MCS0=1*40=40$, $Cost_ETT.sub.MCS1=1*20=20$, $Cost_ETT.sub.MCS2=1*13.333=13.3333$,

$Cost_ETT.sub.MCS3=1*10=10$, $Cost_ETT.sub.MCS4=(1/0.8333)*6.6667=8$, $Cost_ETT.sub.MCS5=(1/0.5)*5=10$, $Cost_ETT.sub.MCS6=$

$(1/0)*4.444=Inf$, $Cost_ETT.sub.MCS7=(1/0)*4=Inf$

$Cost_ETT$

(94) Thus, the overall cost, $cost_ETT$ at **950** is the lowest value of $Cost_ETT.sub.MCS$ over all cases of MCS. Thus $Cost_ETT='8'$.

(95) It is envisaged that the aforementioned example of FIG. 9 can be adapted according to the examples of the concepts herein described, and the specific number and ranges of values are provided for simplicity of illustration purposes only. It is also envisaged that examples of the invention can be applied in a variety of wireless mesh networks, such as WiFi™ 802.11 networks, routing protocols, employing both distance vector and link state. Note that all

used to determine the number of rows in a sounding history array)).

(102) After the ‘a priori’ non-peer specific parameters have been determined the peer node specific parameters are obtained by looking within the information contained in the sounding transmission at **1325**, say, the new sub-TLV. For example, within the sub-TLV, the rate table index and the number of sounding transmissions in the sounding burst, Nburst, are obtained (note that if the non-consecutive MCS option is chosen then in addition to the Nburst parameter an additional MCS bitmap will need to be received). Thereafter, at **1330**, the peer node context is created and distributed at **1340**, **1350**. Here, the sounding history array for the peer node may also be created, where the Nburst parameter is used to configure the number of columns and the sounding history depth used to configure the number of rows. Note that there are multiple instances of this functionality for all the peer nodes which have sounding bursts received by this node. Additionally the MCS rate table history may be used to look up the appropriate rate table that is used to calculate the ETT in the peer node context functionality in FIG. **14**.

(103) Referring now to FIG. **14** an example flowchart illustrates a method showing peer node context sounding burst reception functionality, in accordance with some example embodiments of the invention. The flowchart starts at **1405**. At **1410**, the receiving node of the distributed peer node context starts a second timer T2 and a third timer T3. The second timer T2 controls the rate at which the receiver ETT metric is calculated; this would typically be equivalent to the sounding history depth. The third timer T3 may be used to ensure that when no sounding transmissions within the sounding burst are received the sounding history is updated with all zeros for that period (typically T3 is set to slightly longer than the transmit sounding burst periodicity, i.e. T1 in FIG. **11**). Note that **1410** only needs to be run when the peer entity has first been created. Subsequently the main loop of the functionality (described below) continuously runs.

(104) In the main loop of the peer node functionality there are three decision boxes. A first decision box, at **1420**, determines whether (or not) a sounding burst is received. If, at **1420**, a sounding burst is received then the sounding history is shifted down one row, with the contents of the bottom row discarded and all zeros placed in the top row at **1425**. Thus, effectively, a FIFO structure is maintained. Furthermore, at **1430** and based on the individually received/not received sounding transmissions within the sounding burst, the receiving node fills in the top row of the sounding history utilizing the intra-burst sequence number to identify the appropriate columns. For instance, if individual sounding transmissions with intra-burst sequence numbers 1, 2, 3 and 5 are received then add a ‘1’ to in columns 1, 2, 3 and 5 of the top row in the sounding history array. At **1435**, the timer T3 is re-started (i.e. the timer used to determine that nothing in the sounding burst is received).

(105) In a second decision box at **1440**, a determination is made as to whether (or not) timer T2 has expired. If timer T2 has expired then at **1445**, the peer node calculates an ETT using the current sounding history (say, from the sounding history array) together with the appropriate MCS rate table information (using the mechanism outlined in FIG. **9**). Thereafter, at **1450**, timer T2 is reset and restarted.

(106) In a third

and comprising: a transmitter configured to transmit transmissions to at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between the wireless communication unit and the receiving wireless communication unit; and a processor coupled to the transmitter and arranged to generate a sounding burst comprising: a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS to be transmitted to the plurality of neighboring wireless communication units; and a plurality of parameters contained at an application layer protocol level; wherein the plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst.

2. The wireless communication unit of claim 1, wherein the processor generates a further parameter contained at the application layer protocol level in the plurality multicast sounding transmissions of a sounding burst, wherein the further parameter indicates a respective number of an individual multicast sounding transmission that is used in the sounding burst.

3. The wireless communication unit of claim 2, wherein the number of individual multicast sounding transmissions that are used in the sounding burst is a same number value across the plurality of multicast sounding transmissions.

4. The wireless communication unit of claim 1, wherein the intra-burst sequence number for each MCS transmitted within each multicast sounding transmission contains a non-consecutive MCS in each subsequent sounding transmission of the sounding burst.

5. The wireless communication unit of claim 1, wherein the processor generated sequence number for each MCS transmitted within the plurality of multicast sounding transmissions at the application layer protocol level is generated in a MCS rate table that is transmitted in each multicast sounding transmission.

6. The wireless communication unit of claim 5, wherein the MCS rate table is indexed using a parameter that points to a particular table of a previously stored set of tables.

7. The wireless communication unit of claim 6, wherein the MCS rate table parameter comprises a predefined set of MCS rate tables that uses a relative value between a peak bit rate and an individual bit rate contained in a MCS rate table together with predefined MCS rate table indexes.

8. The wireless communication unit of claim 6, wherein the MCS rate table parameter points to a previously defined 802.11 MCS rate table that enables the at least one neighboring wireless communication unit to calculate an Estimate Transmission Time, ETT, cost metric.

9. The wireless communication unit of claim 1, wherein the processor generates the sounding burst comprising the plurality of multicast sounding transmissions by including a sub Type Length Value, TLV, encoding sounding transmission that includes a legacy burst sequence number.

10. A method for determining a plurality of link costs for communications to a receiving wireless communication unit in a wireless mesh network in a wireless communication unit configured to use a multiple modulation and coding scheme, MCS, the method comprising: transmitting sounding transmissions to at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between the wireless communication unit and the receiving wireless communication unit; and generating a sounding burst comprising a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS to be transmitted to the plurality of neighboring wireless communication units; and a plurality of parameters contained at an application layer protocol level; wherein the plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst, to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst.

11. A receiving wireless communication unit for determining a communication link cost metric from a sounding burst received from a transmitting wireless communication unit received via a plurality of communication links in a wireless mesh network, the receiving wireless communication unit configured to use a multiple modulation and coding scheme, MCS, and comprising: a receiver configured to receive sounding transmissions from at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between the transmitting wireless communication unit and the receiving wireless communication unit; and a processor coupled to the receiver and arranged to receive a sounding burst comprising a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS from respective neighboring wireless communication units of the plurality of neighboring wireless communication units; and each multicast sounding transmission comprises a plurality of parameters contained at an application layer protocol level; wherein the plurality of parameters in the sounding burst comprises: an intra

communication unit configured to use a multiple modulation and coding scheme, MCS, the method comprising: receiving sounding transmissions from at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between a transmitting wireless communication unit and the receiving wireless communication unit; and receiving and processing a sounding burst comprising a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS from respective neighboring wireless communication units of the plurality of neighboring wireless communication units; and each multicast sounding transmission comprises a plurality of parameters contained at an application layer protocol level; wherein the plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst; calculating a cost metric for the respective received sounding burst for each of the received sounding transmissions received via the plurality of communication links.

20. A wireless communication system for determining a plurality of communication link cost metrics comprising a transmitting wireless communication unit for communications to a receiving wireless communication unit in a wireless mesh network, the transmitting wireless communication unit configured to use a multiple modulation and coding scheme, MCS, and comprising: a transmitter configured to transmit sounding transmissions to at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between the transmitting wireless communication unit and the receiving wireless communication unit; and a processor coupled to the transmitter and arranged to generate a sounding burst comprising: a plurality of multicast sounding transmissions, wherein each multicast sounding transmission employs a different MCS to be transmitted to the plurality of neighboring wireless communication units; and a plurality of parameters contained at an application layer protocol level; wherein the plurality of parameters in the sounding burst comprises: an intra-burst sequence number for each MCS transmitted within each multicast sounding transmission; an indication of a number of sounding transmissions in the sounding burst; and an index value that references a set of values that are specific to the wireless configuration that the wireless remote communications unit employs and used together with the intra-burst sequence number that identifies a specific sounding transmission within the sounding burst to determine a relative bit rate value associated with each multicast sounding transmission within the sounding burst; and wherein the receiving wireless communication unit comprises: a receiver configured to receive sounding transmissions from at least two neighboring wireless communication units of a plurality of neighboring wireless communications units, wherein the wireless mesh network comprises the at least two neighboring wireless communication units forming a respective communication path between the transmitting wireless communication unit and the receiving wireless communication unit; and a processor coupled to the receiver and arranged to receive the sounding burst, wherein the processor is configured to calculate a cost metric for a respective received sounding burst for each of the received sounding transmissions received via the plurality of communication links.
